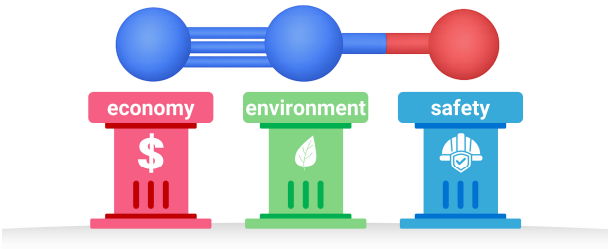

Using economic, environmental, and safety metrics, we demonstrate that nitrous oxide obtained via ammonia oxidation outperforms the conventional five-step route and a well-established oxidant such as hydrogen peroxide, renewing key applications such as the one-step conversion of benzene to phenol.